



SUPPLY CHAIN ANALYTICS

DASHBOARD USING POWER BI

Transforming supply chain data into actionable insights for better decisions, efficiency, and business growth.



“ Connecting data.
Optimizing operations.
Delivering value. ”

TOOLS & TECHNOLOGIES USED



Power BI
Desktop



Power
Query



DAX



Microsoft
Excel



Data
Modeling



CSV
Data Source

PROJECT FOCUS AREAS



Inventory
Management



Supplier
Performance



Logistics
Operations



Warehouse
Analysis



Business
Insights



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SUBMISSION DATE
4th July 2026

1. PROJECT OVERVIEW

Supply Chain Analytics plays an important role in improving operational efficiency by providing visibility into inventory, supplier performance, warehouse operations, logistics, and order management.

This project was developed using Microsoft Power BI to create an interactive dashboard that helps organizations monitor supply chain activities and make informed business decisions using real-time insights.

2. PROJECT OBJECTIVE

The main objective of this project is to design and develop an interactive Supply Chain Analytics Dashboard that enables management to:

- Monitor inventory levels
 - Track supplier performance
 - Analyze logistics operations
 - Monitor warehouse utilization
 - Evaluate delivery performance
 - Improve operational efficiency
 - Support data-driven decision-making
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3. BUSINESS PROBLEM STATEMENT

Supply chain operations involve large amounts of transactional data, making it difficult to identify operational issues through traditional reporting methods.

The organization requires an interactive dashboard to:

- Monitor inventory availability
 - Identify low-stock products
 - Evaluate supplier efficiency
 - Track shipment performance
 - Analyze transportation costs
 - Compare warehouse performance
 - Improve customer delivery performance
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4. TOOLS & TECHNOLOGIES

Data Source

- CSV Dataset

Data Preparation

- Microsoft Power Query

Dashboard Development

- Microsoft Power BI Desktop

Data Modeling

- Star Schema

Calculations

- DAX

Visualization

- KPI Cards
- Charts
- Maps
- Treemap
- Gauge
- Donut Charts
- Slicers

5. DATASET DESCRIPTION

The dataset contains operational information related to the supply chain.

Major Data Fields

- Order ID
- Customer ID
- Customer Name
- Product ID
- Product Name
- Category
- Department
- Order Date
- Shipping Date
- Region
- Country
- City
- Sales
- Profit
- Product Price
- Delivery Status
- Shipping Mode
- Order Status

6. DATA CLEANING PROCESS

The dataset was cleaned using Power Query before loading into Power BI.

Steps Performed

- Removed duplicate records
- Corrected inconsistent values
- Standardized country names
- Standardized city names
- Standardized state names
- Corrected special characters
- Removed unnecessary columns
- Renamed columns
- Corrected data types
- Formatted date columns
- Validated missing values

- Loaded clean dataset into Power BI
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7. DATA MODELING

A Star Schema was implemented for efficient reporting.

Fact Table

- FactOrders

Dimension Tables

- Dim Date
- Dim Product
- Dim Supplier
- Dim Warehouse

Supporting Table

- Fact Inventory

The data model was optimized to improve report performance and ensure proper interaction between visuals.

8. DIMENSION TABLES

The following dimension tables were created using DAX.

Dim Date

Used for

- Year
 - Quarter
 - Month
 - Date analysis
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Dim Product

Contains

- Product
 - Category
 - Department
 - Product Price
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Dim Supplier

Contains

- Supplier Name
 - Supplier Rating
 - Lead Time
 - Supplier Region
 - Performance
-

Dim Warehouse

Contains

- Warehouse Name
- Warehouse Capacity

- Warehouse Status
-

Fact Inventory

Contains

- Current Stock
 - Reorder Level
 - Safety Stock
 - Inventory Value
 - Inventory Status
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9. RELATIONSHIP MODEL

The following relationships were established:

- Dim Date → FactOrders
- Dim Product → FactOrders
- Dim Product → Fact Inventory
- Dim Supplier → Fact Inventory
- Dim Warehouse → Fact Inventory

Cross-filter direction was set to Single to improve performance.

10. DAX MEASURES

The following business measures were created:

Sales KPIs

- Total Sales
- Total Profit
- Total Orders

Product KPIs

- Total Products
- Total Inventory
- Inventory Value
- Low Stock Products

Supplier KPIs

- Total Suppliers
- Average Lead Time
- Best Performing Supplier

Logistics KPIs

- Total Shipments
- Delayed Shipments
- On-Time Deliveries
- Transportation Cost

Warehouse KPIs

- Active Warehouses
 - Total Stock Available
 - Warehouse Performance
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11. DASHBOARD DEVELOPMENT

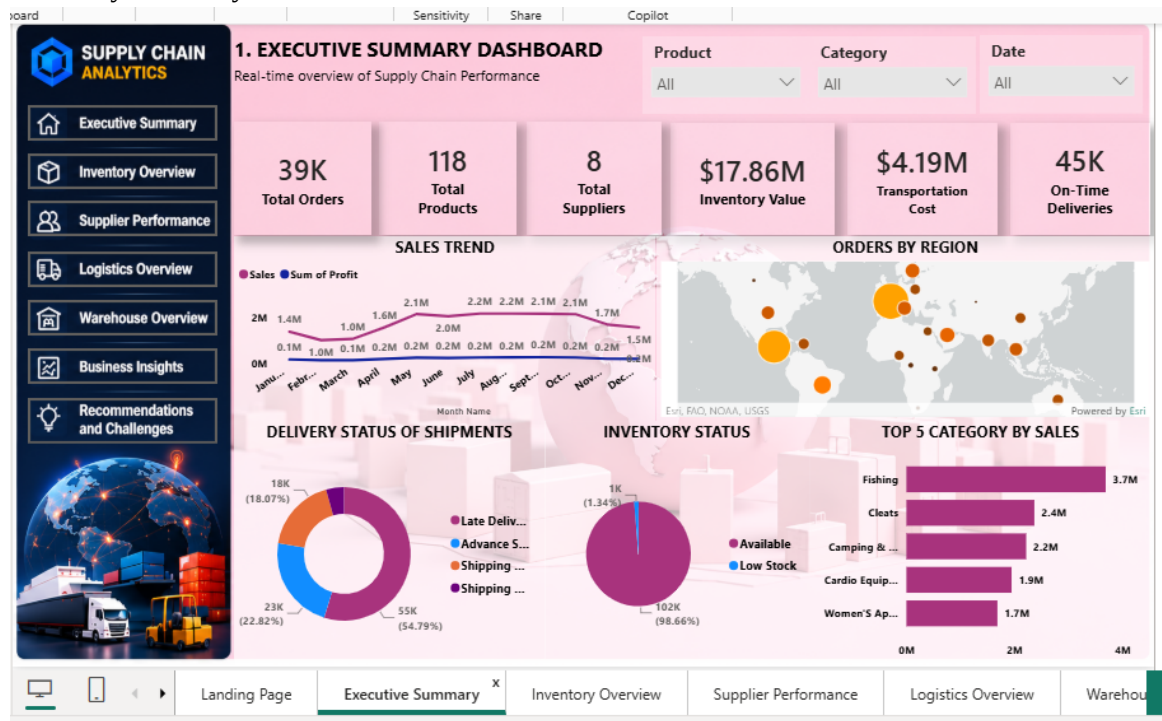
Five analytical dashboards were created.

Dashboard 1

Executive Summary

Displays

- Business KPIs
- Sales Trend
- Regional Analysis
- Delivery Performance
- Inventory Summary

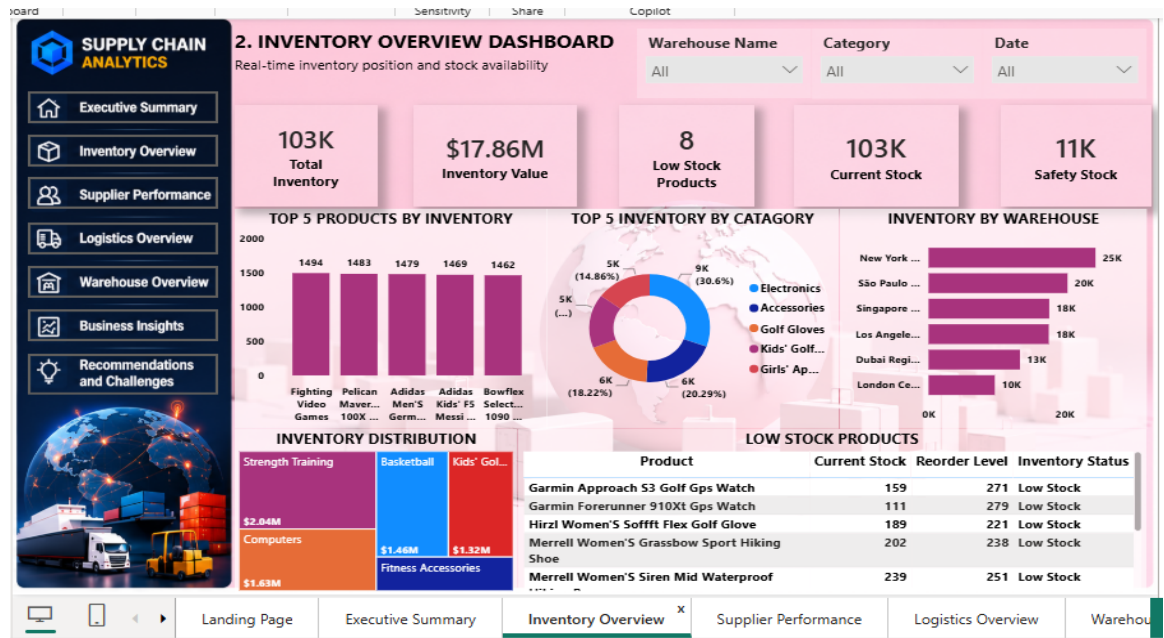


Dashboard 2

Inventory Overview

Displays

- Inventory Value
- Product Inventory
- Warehouse Inventory
- Low Stock Products
- Inventory Distribution

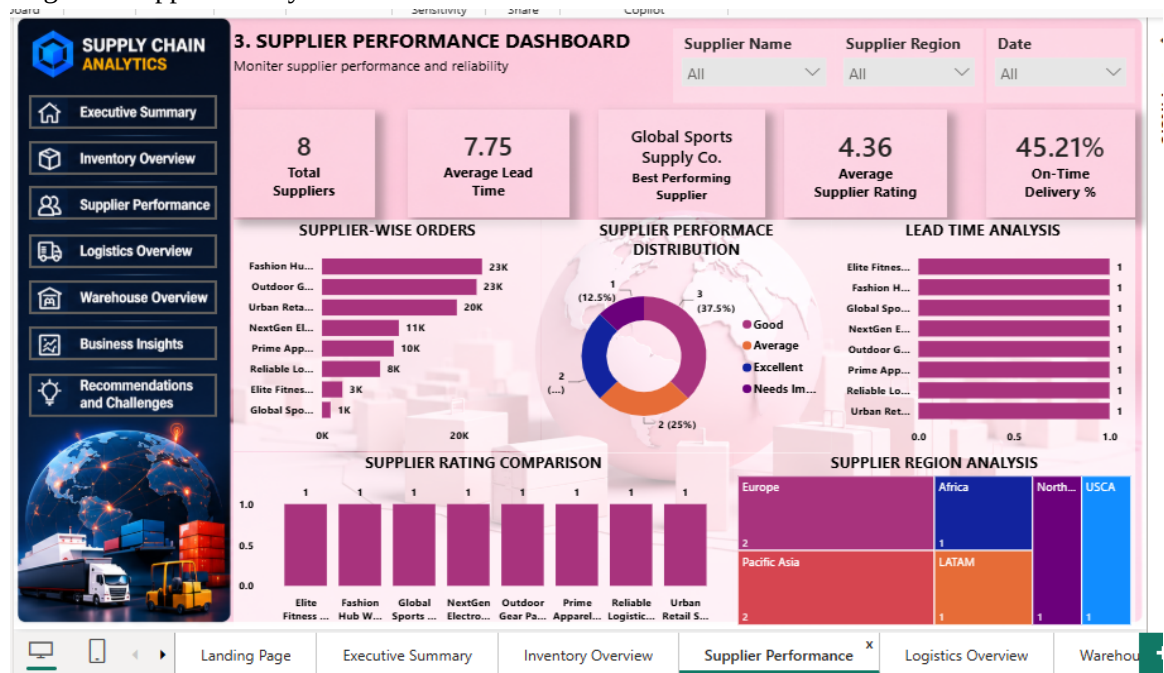


Dashboard 3

Supplier Performance

Displays

- Supplier Ratings
- Supplier Lead Time
- Supplier Orders
- Supplier Performance
- Regional Supplier Analysis

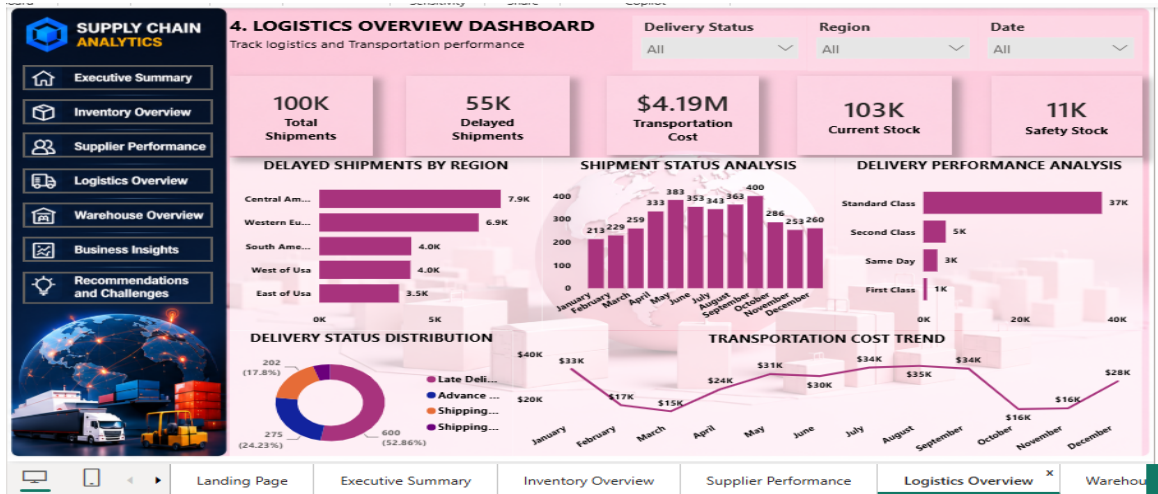


Dashboard 4

Logistics Overview

Displays

- Shipment Analysis
- Delivery Status
- Transportation Cost
- Delayed Shipments
- Shipping Performance

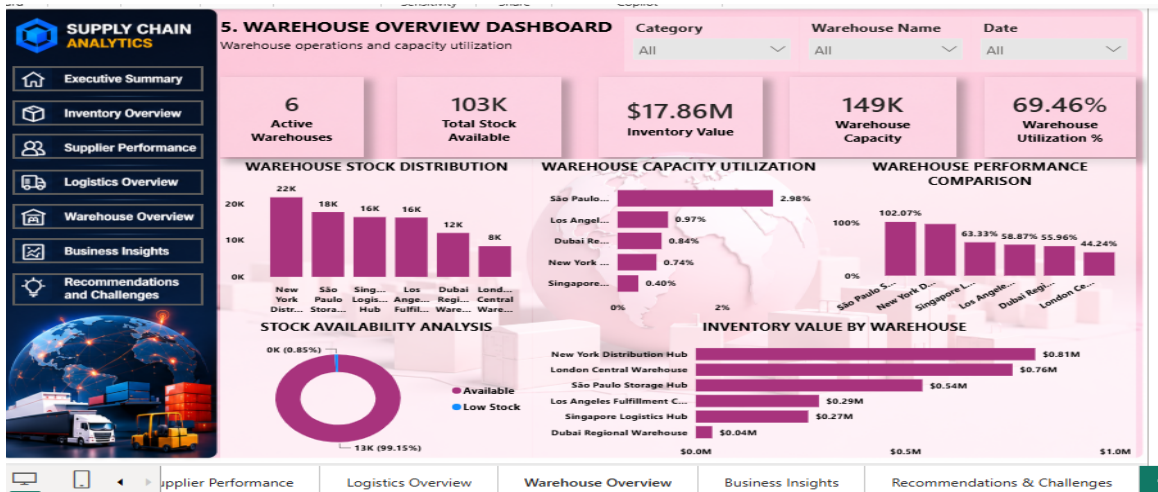


Dashboard 5

Warehouse Overview

Displays

- Warehouse Inventory
- Warehouse Capacity
- Inventory Value
- Stock Availability
- Warehouse Utilization



12. DASHBOARD FEATURES

The dashboard includes:

- Interactive KPI Cards
 - Dynamic Slicers
 - Navigation Buttons
 - Drill Through
 - Bookmarks
 - Dynamic Titles
 - Cross Filtering
 - Professional Theme
 - Interactive Charts
 - Business Insights Page
 - Recommendations Page
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13. BUSINESS INSIGHTS

The dashboard provides valuable insights into:

- Inventory availability
 - Low stock products
 - Supplier performance
 - Warehouse efficiency
 - Shipment delays
 - Transportation costs
 - Regional order distribution
 - Product demand
 - Delivery performance
 - Business trends
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14. RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested:

- Replenish low-stock products promptly.
 - Monitor supplier performance regularly.
 - Improve transportation route planning.
 - Optimize warehouse utilization.
 - Reduce delivery delays through proactive tracking.
 - Maintain balanced inventory across warehouses.
 - Focus on high-demand product categories.
 - Use dashboard insights for continuous operational improvements.
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15. CHALLENGES FACED

- Cleaned duplicate and inconsistent records.
- Standardized country, state, and city names.
- Corrected data types and date formats.
- Designed a Star Schema data model.
- Created relationships between Fact and Dimension tables.

- Developed DAX measures and calculated tables.
 - Built interactive dashboards with navigation.
 - Optimized report performance and visual interactions.
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16. LEARNING OUTCOMES

Through this project, I gained practical experience in:

- Power Query transformations
 - Data cleaning techniques
 - Star Schema modeling
 - Relationship management
 - DAX calculations
 - KPI development
 - Interactive dashboard design
 - Business intelligence reporting
 - Data visualization best practices
 - Business decision support
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17. CONCLUSION

The Supply Chain Analytics Dashboard successfully transformed raw operational data into meaningful business insights. The dashboard provides complete visibility into inventory, suppliers, warehouses, logistics, and order management through interactive visualizations. The project demonstrates practical knowledge of Power BI, Power Query, DAX, data modeling, and business intelligence, enabling organizations to make faster and more informed decisions.